

ASSIGNMENT:2

TCP/IP Suite Theory:

Task (1):

Layer	Example Protocol	What to Look For in Headers
Application	HTTP/HTTPS	HTTP request/response, or TLS information for HTTPS
Transport	TCP	Source port, destination port, sequence number, flags
Internet	IPv4/IPv6	Source IP, destination IP, TTL/Hop Limit
Network Access	Ethernet/Wi-Fi	MAC addresses, frame information

Task (2):

OSI Model vs TCP/IP Model

OSI Model Layer	TCP/IP Model Layer	Main Function	Real-World Example
7. Application	Application	Provides network services directly to applications.	WhatsApp – sending a message or Zomato – placing an order.
6. Presentation	Application	Handles data formatting, encryption, and compression.	WhatsApp – encrypting messages before transmission.
5. Session	Application	Establishes, manages, and terminates communication sessions.	Google Meet – maintaining a video-call session.
4. Transport	Transport	Provides end-to-end data delivery and reliability.	YouTube – transporting video data between the server and your device.
3. Network	Internet	Handles IP addressing and routing between networks.	Zomato – routing your order request from your phone to its servers.
2. Data Link	Network Access	Transfers data between devices on the same local network.	Wi-Fi – sending your phone's data to your home router.
1. Physical	Network Access	Transmits raw bits through physical or wireless signals.	Wi-Fi radio signals or an Ethernet cable carrying network data.

Task (3):

Source Port: identifies which application/process on your device sent the data (e.g. your browser's ephemeral port)

Destination Port: identifies which service on the receiving server should get the data (e.g. port 443 for HTTPS)

Sequence Number: tracks the order of bytes sent, so the receiver can reassemble segments correctly

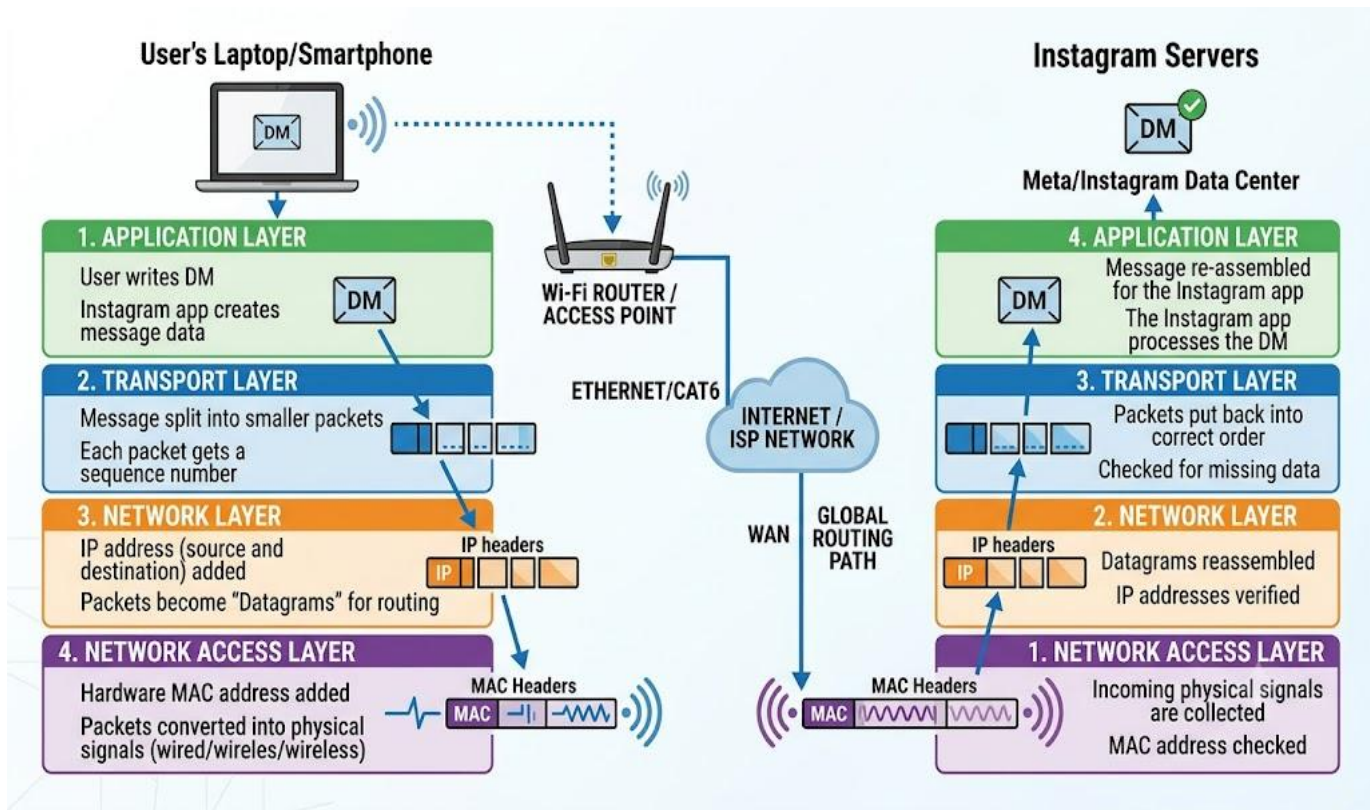
Acknowledgment Number: confirms which bytes have been successfully received so far

Flags (SYN, ACK, FIN, etc.): control bits that manage connection setup, data acknowledgment, and connection teardown

Window Size: tells the sender how much data the receiver can accept before needing another acknowledgment

Checksum: used to detect errors/corruption in the transmitted segment

Task (4):



Task (5):

Item	Answer
Selected Packet	TCP packet
TCP/IP Layer	Transport Layer
How I Identified It	The Wireshark Protocol column shows TCP, and the packet details contain TCP fields such as Source Port, Destination Port, Sequence Number, and Flags .
Real-World Action	It can represent data being transported when I send a WhatsApp message , such as part of the communication between my phone and WhatsApp's servers.